ITCC508 Lab Exercise PT-F1

Optimizing Chunk Size and Overlap — Initial Preps: Code Review, Test Questions, and Prompt Settings

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1. Dataset

The indexed document is **Sagarino_PT-F1_Corpus.pdf**, a 17-page consolidated reference of the four Philippine statutes and the agency guidance that govern remedies for a defective consumer purchase: Republic Act No. 7394 (Consumer Act), No. 8792 (E-Commerce Act), No. 10642 (Philippine Lemon Law) and No. 11967 (Internet Transactions Act), together with the Department of Trade and Industry's published consumer guidance and complaint procedure. It is the corpus of my proposed RAG application.

Statutory text is verbatim. Explanatory notes I had written for my own use were removed before the PDF was produced, so every retrievable passage is primary source material and no test answer is supported by our own commentary.

Check	Result
Extractable text	All 17 pages return text under pypdf. Mean 3,584 characters per page; no page below 409. No OCR required.
Valid FAISS store	Built at the 1000/200 baseline: 85 chunks, 85 vectors, 384 dimensions, IndexFlatIP on normalised vectors. Vector count equals chunk count.
Chunk inspection	Exported to Sagarino_PT-F1_Corpus_chunks_1000_200.xlsx with chunk ID, page, character count, both settings and the chunk text.
Evidence integrity	Each passage supporting a test question was confirmed to fall whole inside a single chunk rather than across a boundary.

2. Test questions, expected answers and evidence

Three questions are required, one at each difficulty level. Difficulty comes from the amount of evidence an answer needs, not from vague wording: all three are fully answerable from the dataset. A fourth item is recorded separately as a guardrail check and is not one of the three graded questions.

ID	Level	Question	Expected answer	Evidence
Q1	Easy	Under the Philippine Lemon Law, how many separate repair attempts for the same complaint must fail before a consumer may invoke their rights under the Act?	Four (4) separate repair attempts.	Sagarino_PT-F1_Corpus.pdf, page 10 - RA 10642, Section 5 (Repair Attempts).
Q2	Average	Under the Internet Transactions Act, what must an aggrieved online buyer do before filing a complaint before any court or government agency, and when is that step deemed exhausted?	The buyer must first use the internal redress mechanism of the digital platform, e-marketplace or e-retailer. It is deemed exhausted if the complaint remains unresolved after seven (7) calendar days from filing.	Sagarino_PT-F1_Corpus.pdf, page 14 - RA 11967, Section 24 (Internal Redress Mechanism).

ID	Level	Question	Expected answer	Evidence
Q3	Difficult	A consumer's brand-new motorcycle has failed four separate repair attempts within six months of delivery. Can they invoke the Philippine Lemon Law? If not, which statute governs and what remedy does it give?	No. Section 3 (j) of RA 10642 defines a motor vehicle as a four-wheeled road vehicle and expressly excludes motorcycles, and Section 4 limits coverage to brand new motor vehicles, so the four-repair rule in Section 5 does not apply. Article 100 of RA 7394 governs instead: if the imperfection is not corrected within thirty (30) days, the consumer may demand replacement of the product, reimbursement of the amount paid, or a proportionate price reduction.	Sagarino_PT-F1_Corpus.pdf, page 9 - RA 10642, Section 3 (j) and Section 4; page 10 - RA 10642, Section 5; page 4 - RA 7394, Article 100.
G1	Guardrail (not graded)	How many days does Shopee give a buyer to request a refund?	I cannot find the answer in the provided document.	Not in the corpus. A marketplace refund window is a commercial term set by contract, not a statutory entitlement, and appears in none of the five sources.

Why Q3 is difficult. It requires three passages across two statutes and an exclusion that must be recognised before the applicable rule can be stated. A system that matches on surface similarity retrieves the Lemon Law's four-repair rule and answers yes, which is wrong. The correct answer has to notice that Section 3 (j) excludes motorcycles from the definition of a motor vehicle, and then move to a different statute entirely.

3. Final Language Model

openai/gpt-oss-20b, served through the Groq API.

This is the model assigned for this exercise. It replaces google/flan-t5-base, which the provided notebook used. Three changes follow from the substitution, all implemented in Component II of the notebook:

Change	Reason
Context budget raised from 512 to 8,000 tokens	Flan-T5 accepts 512 input tokens, so the original prompt builder truncated evidence mid-chunk. Truncated evidence is a confound: it is indistinguishable in the output from a retrieval miss. All three retrieved chunks are now passed whole.
Answer budget raised from 32 to 1,024 tokens, with reasoning effort set to low	gpt-oss-20b is a reasoning model: it spends tokens thinking before it answers, and on Groq those tokens come out of the same completion budget. A first run capped at 320 returned no answer at all for the average and difficult questions, both having used exactly 320 tokens, while the easy and guardrail questions needed only 157 and 187. The larger budget and the low reasoning setting leave room for the answer itself.
Beam search replaced by temperature 0	The same questions are asked once per chunking configuration. A model that varies its wording between runs makes it impossible to attribute a change in the answer to the change in chunking.
Model identifier resolved at run time	Hosted models are deprecated without notice; a hard-coded identifier can begin returning 404 with nothing in the notebook having changed. The notebook queries the API for the served model list and falls back through a ranked order.
API key read through getpass	The key is never written into a cell, printed, or saved, so the notebook can be submitted and shared without exposing a credential.

4. Final Prompt

The prompt is sent in two parts. The **system message** carries the grounding contract and the guardrails; the **user message** carries the question and the retrieved evidence. Separating them is deliberate — an instruction placed in the same block as retrieved text can be read by the model as part of that text.

System message (verbatim, as sent)

```
You are a retrieval-grounded assistant for Philippine consumer-remedy law. You answer only from
the Evidence supplied in the user message.

ANSWER FORMAT - use exactly this shape, including the Scope line:
  <one or two sentences that directly answer the question>
  Supporting evidence: "<verbatim quote from the Evidence>" [Page N]
  Scope: <the Evidence provision that brings the item named in the question within the rule you
applied, quoted and cited - or the words 'not established in the Evidence'>

Rules:
1. Use only the Evidence. Do not use prior knowledge, and do not import rules from the consumer
law of any other country.
2. Read every Evidence block before answering. They may come from different statutes; use the
one that addresses the question asked.
3. Always give the answer sentence first. A quotation on its own is not an answer. Then give the
supporting quote, reproduced word for word from the Evidence, with the page cited immediately
after it. Never cite a page you did not quote from.
4. Scope check. Before stating that a rule applies to the item named in the question, confirm
that some Evidence block contains the definition or coverage provision that brings that item
within the rule. If no block defines or lists the item, you must not assume it is covered:
answer The evidence does not establish whether this item is covered. and name the definition or
coverage provision that is missing. A broad category in the Evidence does not automatically
include a narrower item named in the question. If the question names a specific kind of thing
and the Evidence only uses a wider category, you must find the provision that defines that
category before treating the item as falling inside it.
5. If the Evidence shows that a rule does not cover the item, say so first, quote and cite the
provision that excludes it, then give the provision that does apply.
6. Refuse only as a last resort, and only after checking every Evidence block. If, having read
them all, none contains the answer, reply with exactly this sentence and nothing else: I cannot
find the answer in the provided document. Do not refuse merely because you are unsure which
block is the right one, or because the question uses different wording from the Evidence.
7. A platform's own refund window, a seller's terms, or any commercial policy is not law. If the
question asks for one, apply rule 6.
8. Quote figures exactly as they appear: days, months, kilometres, number of repair attempts,
amounts.

Be concise. Do not restate the question.
```

User message template

```
Question: {question}

Evidence:
[Page {page}, Chunk {chunk_id}, similarity {score}]
{chunk_text}

[Page {page}, Chunk {chunk_id}, similarity {score}]
{chunk_text}
... one block per retrieved chunk, in descending similarity order ...

Answer:
```

Guardrails and why each one is there

Guardrail	Failure it prevents
Answer only from the Evidence	The model is trained largely on English-language sources, which carry a great deal of American and European consumer law. The EU’s fourteen-day right to withdraw from a purchase has no Philippine counterpart but is the most likely thing an ungrounded model would produce here.
Cite the Act, the Article or Section, and the page	An untraceable answer cannot be verified. The citation also separates a correct remedy attributed to the right statute from one attributed to the wrong statute — the second sends a consumer to the wrong forum on the wrong timetable.
One canonical refusal string	A fixed sentence makes refusals countable. Free-form refusals vary between runs and cannot be scored.

Guardrail	Failure it prevents
Platform policy is not law	A marketplace refund window is a commercial term that appears nowhere in the corpus. Without the rule, the model is liable to supply a plausible number.
Quote figures exactly	Thirty days, twelve months, twenty thousand kilometres, four repair attempts, seven calendar days. These figures are the answers; an approximation is a wrong answer.
Name the exclusion before the applicable rule	The hardest questions in this domain are near-misses, where a rule looks applicable and is not. Requiring the excluding provision to be named converts a silent mistake into a visible step that can be checked.
Quote before concluding	The model must reproduce verbatim the clause it relies on and attach the citation to that quote. This is the strongest single guardrail in the set: a quote either appears in the retrieved text or it does not, which is far harder to fake than a page number.
Confirm scope before applying a rule	Before stating that a rule applies to the item in the question, the model must check that the evidence shows the item is within that rule's scope. If it does not, the model says so instead of assuming. This is what separates a cautious answer from a confident wrong one on a near-miss question.
Report finish_reason on every call	An API error, a truncated completion and a refusal are three different results and must not be scored alike. Reporting finish_reason makes a truncation name itself instead of appearing as an empty answer.
Quote verification (in code)	The notebook reads the quoted passages back out of the answer and confirms each one occurs verbatim in the retrieved chunks, alongside the older check that every cited page was retrieved. The page check alone proved too weak: it passed an answer that cited a retrieved page while describing a procedure that appears nowhere in the statute.

5. Configuration held fixed for the experiment

PT-F1 prepares a controlled comparison of chunk size and overlap. Everything below is held constant so that a difference in Precision@3, Recall@3 or MRR@3 is attributable to the chunking configuration and to nothing else.

Setting	Value	Status
Corpus	Sagarino_PT-F1_Corpus.pdf, 17 pages	fixed
Embedding model	all-MiniLM-L6-v2, 384 dimensions, normalised	fixed
Index	FAISS IndexFlatIP (exact; cosine on unit vectors)	fixed
Top-K	3 — the metrics are defined at 3	fixed
Language model	openai/gpt-oss-20b via Groq	fixed
Temperature	0.0	fixed
Max new tokens	1,024	fixed
Reasoning effort	low (gpt-oss family)	fixed
Prompt and guardrails	as printed in section 4	fixed
Test questions	Q1, Q2, Q3 plus guardrail G1	fixed
Chunk size	1,000 characters at baseline	**varied**
Chunk overlap	200 characters at baseline (20%)	**varied**

One property of the provided chunker worth recording. `create_chunks` restarts at offset 0 on every page, so a page break is always a hard chunk boundary and overlap never bridges two pages. A passage split across a page break is therefore split across chunks at every setting. This is a property of the implementation, not of the parameters, and it should be kept in mind when interpreting results: a question whose evidence straddles a page break is harder for a reason unrelated to the chunk size under test.

6. How the prompt reached its final form

The prompt in section 4 is the third version. Each earlier one was replaced because a run exposed a specific failure, and recording that sequence is more useful than presenting the final prompt as though it had been obvious.

Version	What the run showed	What changed
1	The average and difficult questions returned no answer at all. Both had used exactly 320 completion tokens, the configured cap, while the easy and guardrail questions used 157 and 187 and were unaffected.	Identified as truncation, not an empty response: gpt-oss-20b spends tokens reasoning before it answers. Budget raised to 1,024, reasoning effort set to low, and finish_reason now reported so a truncation names itself.
2	All four questions answered. But Q2 described a "DTI-mandated dispute-resolution process" that appears nowhere in Section 24, and cited page 10 — the Lemon Law — although the correct passage on page 14 had been retrieved and was in the prompt. Q3 answered a confident "Yes" on evidence that never contained the exclusion. The grounding check passed Q2 as grounded, because the page it cited had in fact been retrieved.	Two prompt rules added: quote the clause relied on verbatim and attach the citation to that quote; and confirm an item is within a rule's scope before applying the rule. The grounding check was extended to verify that each quoted passage occurs in the retrieved text, which is what the page check alone could not do.
3	Q1 and G1 behaved. But Q2 now *refused* a question whose answer had been retrieved at rank 2 and was sitting in the prompt, and Q3 still answered "Yes". The hardened rules had traded one failure for its opposite: fabrication became over-refusal. Testing the checker also exposed a defect in the checker itself — it searched only for straight quotation marks, so a correctly quoted answer that used typographic quotes was recorded as having quoted nothing.	An explicit answer format was imposed (answer sentence first, then the supporting quote), refusal was restricted to a last resort after every block has been read, and the scope rule was sharpened to require that a definition or coverage provision actually be present. The quote extractor now normalises quotation marks before matching.
4	Q1 answered correctly in the required format and its quote verified. Three problems remained, and two of them were in the checker rather than in the model. G1 declined correctly but in its own wording, and was scored as ungrounded because the check matched only the canonical sentence. Q3 abridged a genuine quotation with an ellipsis and was reported as having quoted something not in the evidence. Q2 still refused a question whose answer had been retrieved — its wording named the DTI, a term Section 24 does not use, and the model would not bridge the gap.	Declining is now detected by pattern rather than by exact wording, so a correct refusal in the model's own words counts as grounded. An abridged quote is verified fragment by fragment. The scope rule gained the principle that a broad category in the evidence does not automatically contain a narrower item named in the question. Q2 was restated in the statute's own terms, since the exercise asks that difficulty come from the evidence required and not from wording that does not match the source.
5	Q1, Q2 and G1 all correct. The reworded Q2 was answered accurately, with the seven-day period and the internal redress mechanism both stated, and its abridged quotation verified fragment by fragment. G1 declined in its own words and was scored correctly this time. Q3 still answered "Yes" — and now did so while quoting the evidence faithfully, so every check that looks only at the model passed it.	A Scope line was added to the required answer format, forcing the model to name the provision that brings the item within the rule or to write that it is not established. The results table gained a verdict that combines grounding with evidence coverage, so an answer that is grounded on partial evidence is reported as UNSAFE rather than appearing to pass.
6	Current version, in section 4.	—

The second iteration is the one worth drawing attention to, because the failure it exposed was in the *checking* rather than in the answer. An inspection that only confirms the cited page was retrieved will pass an answer that cites a retrieved page while asserting something that page does not say. Requiring a verbatim quote and then verifying that quote against the retrieved text closes the gap, because a quote either occurs in the evidence or it does not.

Where the baseline stands

Item	Coverage	Grounded	Verdict
Q1 Easy	1 of 1	yes	SAFE — grounded on complete evidence

Item	Coverage	Grounded	Verdict
Q2 Average	1 of 1	yes	SAFE — grounded on complete evidence
Q3 Difficult	**1 of 3**	yes	**UNSAFE — confident answer on partial evidence**
G1 Guardrail	n/a	yes	SAFE — declined

Q3 is the result worth dwelling on. The answer quotes the Lemon Law accurately and cites a page that was genuinely retrieved, so it passes every check that inspects the model alone. It is still wrong, because the provision excluding motorcycles was never placed in front of the model. This is why the verdict column exists: an answer grounded on one of the three passages a question requires is not a safe answer, and reporting it as grounded without qualification would be the most misleading thing this notebook could do.

Retrieval at the baseline configuration

Question	Evidence pages needed	Retrieved at k = 3	Coverage
Q1 (Easy)	10	10, 10, 1	1 of 1
Q2 (Average)	14	10, 14, 14	1 of 1
Q3 (Difficult)	9, 10, 4	10, 10, 10	**1 of 3**
G1 (Guardrail)	none	12, 3, 12	n/a — refused

Q3 is under-retrieved at this configuration. It needs three passages — the exclusion of motorcycles in Section 3 (j) on page 9, the four-repair rule on page 10, and Article 100 of RA 7394 on page 4. At a chunk size of 1,000 characters the top three results are all drawn from page 10, because the Lemon Law chunks resemble the question most strongly and saturate the available depth. The two passages that would show the rule does *not* apply are never seen.

This is a **retrieval failure, not a generation failure**, and no change to the prompt can repair it: the model cannot reason from evidence it was never given. It is reported here rather than hidden, for two reasons. First, an evidence check that asked only whether *any* expected page was retrieved would have returned a pass for Q3 and concealed the problem; the notebook therefore reports coverage as a fraction. Second, this is precisely the condition the chunk-size experiment exists to resolve, so identifying it during the initial preparation is the intended outcome of this stage rather than a defect in it.

The notebook includes a retrieval diagnostic that searches to a depth of 15 — for diagnosis only, with the scored metrics left at 3 — and reports the rank at which each expected page first appears. That distinguishes a passage sitting just outside the scored depth, which a different chunk size may lift in, from one ranked far below, which no boundary adjustment will recover. At the baseline it measured:

Question	Expected page	First rank	Reading
Q1	10	1	inside the scored depth
Q2	14	2	inside the scored depth
Q3	10	1	inside the scored depth
Q3	9	**4**	**one place outside** — a different chunk size may lift it in
Q3	4	11	far outside — unlikely to be recovered by resizing alone

Three of the five required passages sit inside the scored depth. The most useful single number here is the rank of page 9: the motorcycle exclusion is the *fourth* result, one place below the cut. That is the specific, measured reason to expect the chunk-size experiment to change the outcome on Q3, and it was obtained before the experiment began rather than after it.